Project 42: Terraform Smart Context MCP Server -- Progress Report

Course: CS 6650 -- Distributed Systems, Northeastern University Vancouver
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Repository: github.com/KalharPandya/terraform-smart-context-mcp-42
Project Board: GitHub Projects -- Project 42

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1. Problem, Team, and Overview of Experiments

1.1 The Problem

Large Language Models are increasingly used as infrastructure operators, writing Terraform configurations, debugging deployments, and planning changes. However, when an LLM needs to understand what is *actually deployed*, it faces three specific failures:

1. **State is too large for context windows.** Running `terraform show -json` on a 75-resource project produces 4,041 lines (~33,000 tokens). A real production environment with 500+ resources produces state files that exceed any model's effective context window.
2. **Every interaction starts blind.** The agent has no memory of what is deployed, no way to inspect running services, and no access to filtered views. Each session begins from zero, requiring full state exploration.
3. **Agents guess instead of looking.** Without structured access to live infrastructure, LLMs hallucinate resource arguments, generate deprecated attributes, and produce configurations that fail against real cloud APIs.

The analogy: Code editing agents (Cursor, Claude Code) solved this for programming by giving the AI structured tools like `read_file`, `edit_file`, `run_terminal`. Terraform has no equivalent. There is no standard way for an agent to ask "what's deployed?" or "show me this resource's config" without getting the entire state.

Stakeholders: DevOps teams managing infrastructure at scale, platform engineering teams building internal developer platforms, and any organization using Terraform with 50+ resources where per-query cost and latency matter.

1.2 The Team

Member	Role	Brings to Team
Kalhar Pandya	Technical prototyping, MCP implementation, architecture	Full-stack systems experience, TypeScript/Node.js, MCP SDK integration
Vinal Dsouza	AI-first development workflow, team alignment, project structure	Process design, Claude Code automation workflows, team coordination
Parin Shah	Protocol design, prompt engineering, tool naming, experiments	Experiment design, scorer implementation, baseline data collection

All members are Master's in Computer Science students at NEU Vancouver.

1.3 Overview of Experiments

We designed three experiments to evaluate whether purpose-built MCP tools improve LLM interaction with Terraform infrastructure:

Experiment	Question	Status
1. Tool-Augmented vs. Raw State	Do MCP tools reduce token cost while maintaining accuracy?	Baseline complete, MCP arm pending
2. Abstraction Granularity	What is the optimal granularity: coarse (full dump) vs. fine (filtered)?	Designed, not started
3. Cross-Model Portability	Does the MCP server add value across Claude, GPT, and Gemini?	Designed, not started

Metrics evaluated across all experiments:

> **Context Efficiency** = Total Tokens Used / Correct Answers

Metric	What It Measures
Accuracy	Correctness of agent answers (scored 0.0 / 0.5 / 1.0)
Token Usage	Input + output tokens per query
Cost	USD per query (tokens x rate)
Tool Calls	Number of tool invocations per query
Latency	Wall-clock time per query

1.4 Role of AI

AI plays a **dual role** in this project:

- **AI as development tool:** The entire team uses Claude Code with a structured AI-first workflow (7 slash commands, 5 domain skills) for development, code review, and decision tracking. See Section 4.
- **AI as experiment subject:** Claude (and later GPT and Gemini) is the agent being tested. We measure how its performance changes when given structured MCP tools vs. raw CLI access.

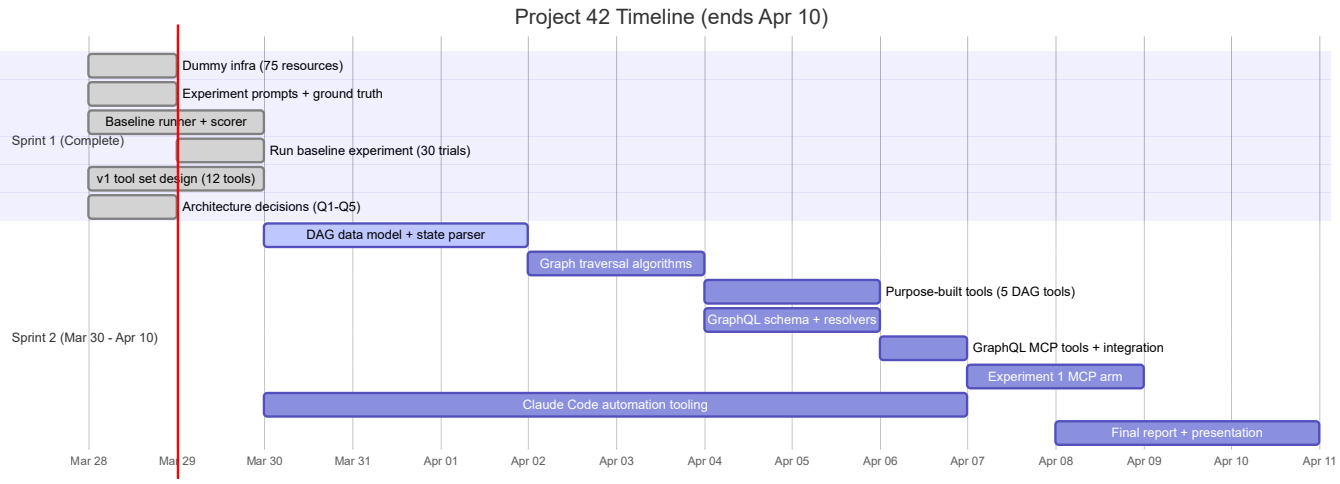
1.5 Observability Plan

Layer	What We Track	How
Experiment	Tokens, cost, time, tool calls, accuracy per trial	Built into `runner.ts`, captures all metrics from Claude Code CLI stream
MCP Server (planned)	Per-tool latency, response size, error rates	Structured JSON logging per tool invocation
Visualization	6-chart interactive dashboard	`visualize.ts` generates Chart.js HTML from scored results
Cost Monitoring	Per-prompt and aggregate cost tracking	CLI `--max-budget-usd` guard + post-hoc cost analysis

2. Project Plan and Recent Progress

2.1 Timeline

Project deadline: **April 10, 2026**. Two sprints total.



2.2 Recent Progress (Sprint 1 -- Complete)

Deliverable	Issue(s)	Owner	Status
75-resource dummy Terraform infrastructure	#12	Parin	Done
10 experiment prompts with ground truth	#13	Parin	Done
Experiment runner (CLI headless mode)	#14	Kalhar, Parin	Done
Experiment scorer (4 scoring types)	#15	Parin	Done
Chart.js visualization pipeline	--	Parin	Done
Baseline experiment (30 trials, $1.73)	#16	Parin	Done
v1 tool set design (12 tools)	--	Kalhar	Done
5 architecture decisions (Q1-Q5)	#7-#11	Kalhar	Done
GitHub issues #17-#23 for implementation	--	Kalhar	Done

2.3 Who Is Doing What

Member	Sprint 1 (Done)	Sprint 2 (Current)
Kalhar	Tool set design, architecture decisions	DAG engine, purpose-built tools, GraphQL, integration (#17-#23)
Vinal	AI workflow design, project structure	Claude Code automation tooling (#1-#6), testing, documentation
Parin	Baseline experiment end-to-end (#12-#16)	Experiment 1 MCP arm, scorer refinements, final experiments

2.4 AI Cost and Benefits in Development

Cost:

- Baseline experiment: **$1.73** for 30 trials (1.49M tokens total)
- Development sessions: Claude Code API costs during iterative development (~$5-10 estimated across Sprint 1)

Benefits:

- A 3-person team shipped the full experiment infrastructure (dummy infra, runner, scorer, visualizer), 12-tool architecture design, and 5 resolved architecture decisions in **one sprint** using Claude Code-first development
- AI-first code review workflow catches decision violations before commit (zero post-merge decision conflicts)
- Automated decision capture ensures all 12 decisions in `DECISIONS.md` are immediately visible to all team members

3. Project Plan (Detailed)

The full project plan is tracked via [GitHub Projects -- Project 42]() with 23 issues across 2 sprints.

3.1 Implementation Phases (#17-#23)

Phase	Issue	Description	Dependencies
1	#17	DAG data model, state parser, precomputed indexes	None
2	#18	Graph traversal algorithms (BFS, path, impact, topo sort)	Phase 1
3	#19	Purpose-built tools: `list_resources`, `get_resource`, `count_resources`	Phase 2
4	#20	Purpose-built tools: `get_dependencies`, `filter_resources`	Phase 2
5	#21	GraphQL schema, resolvers, validation guards	Phase 2
6	#22	GraphQL MCP tools: `query_graph`, `get_schema`	Phase 5
7	#23	Integration: wire 12 tools into `index.ts`, extract CLI, drop `state_list`	Phases 3-6

3.2 v1 Tool Set (12 tools)

#	Tool	Category	What It Returns
1	`list_resources`	DAG	Resources filtered by module/type, summaries only, never full state
2	`get_resource`	DAG	One resource's config + direct neighbors
3	`get_dependencies`	DAG	Dependency subgraph (depth 1-3)
4	`filter_resources`	DAG	Resources matching attribute conditions
5	`count_resources`	DAG	Counts grouped by module/type/tag
6	`query_graph`	GraphQL	Execute GraphQL query against the DAG
7	`get_schema`	GraphQL	Return GraphQL SDL + example queries
8-12	`terraform_*`	CLI	init, validate, plan, apply, output

4. Claude Code-First Development

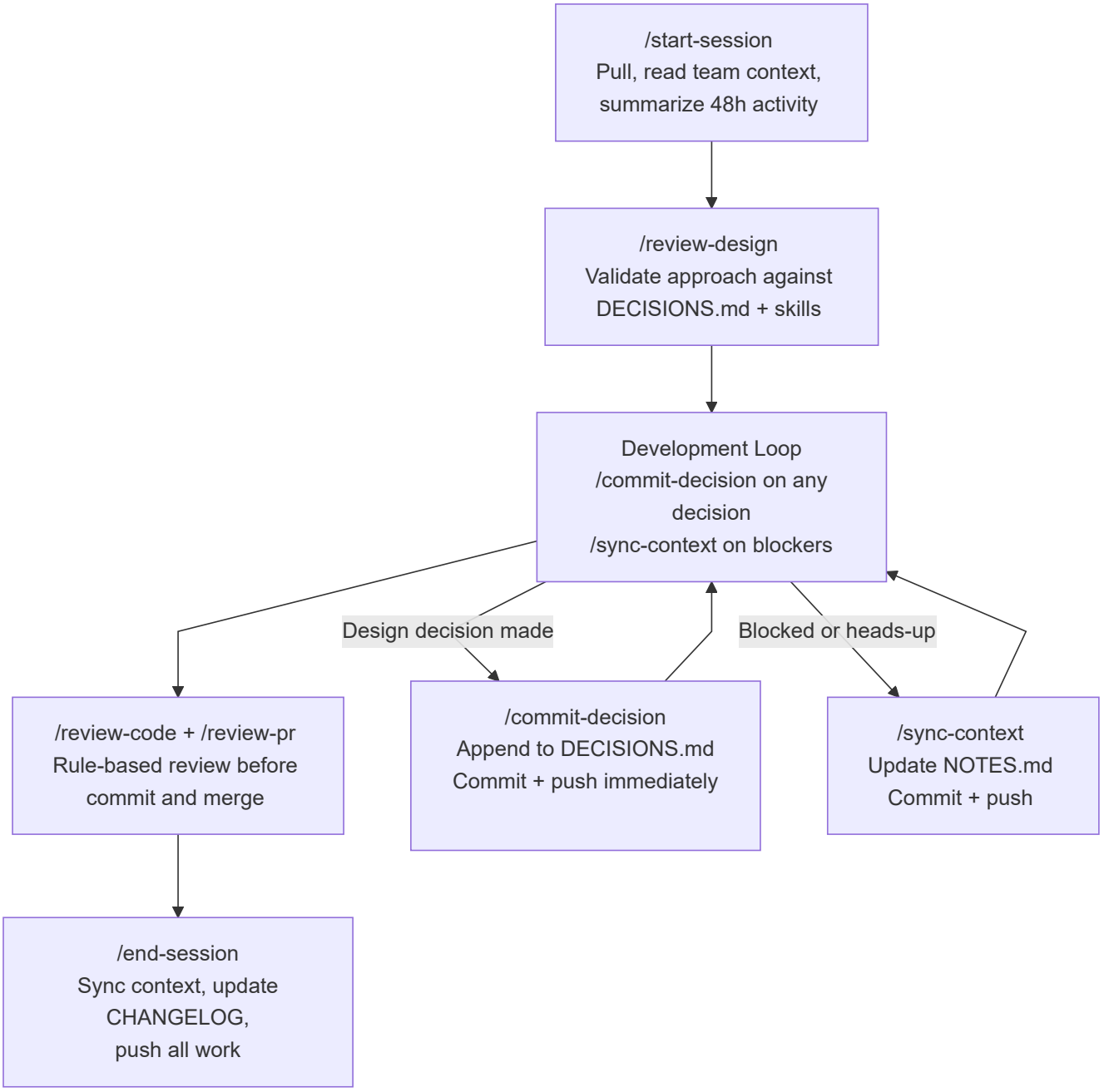
We are experimenting with **Claude Code-first development cycles**, a structured AI-first workflow where Claude Code is not just a coding assistant but the primary interface through which the team develops, reviews, and ships code.

4.1 Automation Inventory

Type	Count	Examples
Slash Commands	7	`/start-session`, `/commit-decision`, `/sync-context`, `/review-design`, `/review-code`, `/review-pr`, `/end-session`
Domain Skills	5	`terraform-parsing`, `dag-design`, `mcp-server-patterns`, `context-reduction`, `code-review`
Config	1	`settings.local.json` (permission rules)
Total	**13 automation files**	All in `.claude/` directory

4.2 Session Lifecycle

Every development session follows a 5-phase cycle enforced by slash commands:



4.3 Key Design Principles

- **Rule-based review, not subjective feedback:** Every review finding must cite a `DECISIONS.md` entry or skill convention. No opinion-based suggestions.
- **Immediate decision capture:** `/commit-decision` runs at the moment of decision, not at end of session. Teammates see it on next `git pull`.
- **Context hierarchy:** `git log` > `DECISIONS.md` > `NOTES.md`. When sources conflict, trust the higher-ranked one.
- **Domain skills auto-load:** When working on `.tf` files, the `terraform-parsing` skill loads automatically. When reviewing code, the `code-review` skill provides the 6-category review framework. No manual invocation needed.

4.4 Insight

The AI is not autonomous. The human makes design decisions; Claude Code documents them immediately, enforces them during review, and broadcasts them to teammates. **Context sharing is the priority, not coding speed.** This workflow treats teammates' visibility of decisions and blockers as higher priority than individual developer velocity.

5. Objectives

5.1 Short-Term (Course Timeline, by April 10)

- **v1 MCP server** with 12 tools (5 DAG + 2 GraphQL + 5 CLI) serving filtered subgraphs from Terraform state
- **6x token reduction** on baseline experiment prompts ($0.058 per query down to ~$0.012)
- All 10 experiment prompts answerable via purpose-built tools with correct ground truth
- Complete Experiment 1 (raw vs MCP comparison) with statistical significance (3+ trials per prompt)

5.2 Long-Term (Beyond Course)

- **Cross-model portability:** Validate that the same MCP tools work across Claude, GPT, and Gemini (Experiment 3)
- **Real cloud support:** v2 parses raw HCL files (not just `terraform show -json` state), enabling pre-apply analysis
- **Open-source release:** Publish as an installable MCP server for the Terraform community
- **CI/CD integration:** MCP server as a pipeline step where agents query infrastructure state during deployment review

5.3 Observability and Reliability

Concern	Mechanism	Detail
Performance	DAG caching	Parse state once on first tool call, cache in-memory; rebuild only after `terraform_apply`
Reliability	Guard limits	GraphQL: depth <= 3, node count <= 50, 5s execution timeout
Cost Control	Token efficiency	Every tool response includes a `summary` string; purpose-built tools return filtered views, not full state
Monitoring	Structured logging	Planned: JSON log per tool invocation with `{tool, latency_ms, response_size, error}`
Experiment Tracking	Per-trial metrics	Runner captures tokens, cost, time, tool calls, answer per trial

5.4 Cost Control for Stakeholders

At baseline, a complex infrastructure query costs **$0.10+** and **79K tokens**. With MCP tools, we project:

> **Cost reduction:** $0.058 / $0.012 = ~5x cheaper per query
>
> **Token reduction:** 49,732 / 8,000 = ~6x fewer tokens per query

For an organization running 100 agent queries/day, this translates from ~$5.80/day to ~$1.20/day, a $1,679/year saving on a single project.

6. Related Work

6.1 Model Context Protocol

Anthropic open-sourced MCP in November 2024 as a standard for connecting AI assistants to external data sources and tools [1]. The protocol uses JSON-RPC 2.0 over stdio or SSE transport, defining three server primitives: Prompts, Resources, and Tools [2]. Early adopters include Block, Apollo, Zed, Replit, and Sourcegraph [3]. Our project implements an MCP server specifically for Terraform infrastructure, a domain not covered by existing MCP servers.

6.2 LLM Tool Use and Function Calling

Research shows that structured tool access significantly reduces LLM hallucination compared to raw text generation. Gorilla [4] demonstrated that LLMs fine-tuned for API calling surpass frontier models at generating correct API calls. The Berkeley Function Calling Leaderboard (BFCL) [5] is the standard benchmark for evaluating tool-calling accuracy across serial, parallel, and multi-turn scenarios, the same patterns our MCP tools target. ToolACE [6] showed that even 8B-parameter models achieve state-of-the-art tool use with well-structured interfaces, validating that **tool design matters more than model size**.

6.3 Infrastructure as Code

Terraform uses an internal directed acyclic graph (DAG) for dependency resolution [7], the same graph structure our MCP server parses and serves to LLMs. TerraFormer [8] demonstrated a 15-20% improvement in LLM-driven Terraform code generation using structured feedback, validating the active research interest in LLM-IaC integration.

6.4 Context Window Limitations

"Lost in the Middle" [9] showed that LLMs exhibit a U-shaped attention curve: they attend well to the beginning and end of context but degrade on information in the middle. "Context Rot" [10] tested 18 frontier models and found **every model degrades well before hitting its maximum context window**, with a 200K-token model showing significant degradation at 50K tokens. This is the core empirical justification for our approach: return minimal, targeted subgraphs rather than dumping entire state.

6.5 Distributed Systems Concepts

MCP directly embodies several distributed systems fundamentals from this course:

Concept	How It Appears
RPC	MCP is built on JSON-RPC 2.0. Each tool invocation is a remote procedure call with typed parameters and return values
Client-Server	The AI application (Claude, Cursor) is the client; our MCP server exposes tools, resources, and prompts
Protocol Design	MCP defines versioned schemas, capability negotiation, and typed message formats, analogous to IDLs
State Management	The server maintains a parsed DAG in memory as server-side state, with session initialization handshakes

6.6 HashiCorp terraform-mcp-server

HashiCorp released their own MCP server for Terraform, but it serves a fundamentally different purpose:

	HashiCorp MCP	Project 42
Answers	"What arguments does `aws_instance` accept?"	"What is deployed in MY environment?"
Data source	Terraform Registry API (documentation)	Live Terraform state + CLI
Dependency graph	No	Yes (core feature)
Filtered views	No	Yes (purpose-built tools)

HashiCorp's MCP is a documentation tool. Ours is an infrastructure introspection tool. They do not overlap in problem space.

References

[1] Anthropic, "Introducing the Model Context Protocol," Nov 2024. https://www.anthropic.com/news/model-context-protocol
[2] Model Context Protocol Specification, 2025. https://modelcontextprotocol.io/specification/2025-11-25
[3] S. De Simone, "Anthropic Publishes Model Context Protocol Specification," InfoQ, Dec 2024.
[4] S. Patil et al., "Gorilla: Large Language Model Connected with Massive APIs," arXiv 2023.
[5] S. Patil et al., "The Berkeley Function Calling Leaderboard," ICML 2025.
[6] Team-ACE, "ToolACE: Winning the Points of LLM Function Calling," Sep 2024.
[7] HashiCorp, "Terraform Resource Graph." https://developer.hashicorp.com/terraform/internals/graph
[8] P. Jana et al., "TerraFormer: Automated IaC with LLMs," Jan 2026.
[9] N. Liu et al., "Lost in the Middle: How Language Models Use Long Contexts," TACL 2024.
[10] Chroma Research, "Context Rot: How Increasing Input Tokens Impacts LLM Performance," 2025.

7. Hypothesis

7.1 Primary Hypothesis

Purpose-built MCP tools that serve filtered, pre-processed subgraphs from Terraform state will reduce LLM token consumption by 5-6x while maintaining or improving task accuracy, compared to raw CLI access.

7.2 Original Predictions (Pre-Experiment)

We predicted that raw CLI performance would degrade sharply on hard tasks:

Difficulty	Predicted Accuracy	Predicted Avg Tokens	Predicted Tool Calls
Easy	80-100%	5-15K	2-4
Medium	50-75%	15-30K	5-8
Hard	10-30%	30-50K	8-15

The reasoning: hard infrastructure questions (dependency chains, impact analysis, deployment order) force iterative state exploration that floods the context window, causing accuracy to collapse.

7.3 Revised Hypothesis (Post-Baseline)

The baseline experiment **partially confirmed and partially surprised us**:

Difficulty	Predicted Accuracy	Actual Accuracy	Predicted Tokens	Actual Tokens
Easy	80-100%	**83%** (confirmed)	5-15K	**28K** (higher than predicted)
Medium	50-75%	**75%** (confirmed)	15-30K	**44K** (higher than predicted)
Hard	10-30%	**94%** (not confirmed)	30-50K	**79K** (much higher)

What we got right: Token cost scales with difficulty (2.8x from easy to hard). Medium accuracy matches prediction.

What surprised us: Hard prompts scored *higher* than medium (94% vs 75%). Claude Sonnet handles multi-hop reasoning well even with raw CLI tools, but at enormous token cost.

Revised thesis: At 75-resource scale, the raw CLI problem is **cost, latency, and context inefficiency**, not accuracy degradation. However, at real-world scale (500+ resources), context overflow would force truncation and *then* accuracy would degrade. The MCP tools solve both: reduce cost now, prevent accuracy collapse at scale.

7.4 MCP Tool Hypotheses (To Be Tested)

Hypothesis	How We Test It	Expected Result
MCP tools reduce tokens per query by 5-6x	Same 10 prompts, raw vs MCP arms	49K avg down to ~8K avg
MCP tools reduce tool calls by 3x	Count tool invocations per prompt	3.5 avg down to ~1.2 avg
MCP tools eliminate token variance	Compare trial-to-trial variance	High (2-3x) down to near-zero (deterministic responses)
MCP tools fix Prompt 3 (scored 0.00 in baseline)	`get_dependencies("vpc", direction: "dependents", depth: 1)`	Exact set returned, no ambiguity
MCP tools collapse Prompt 6 cost ($0.125 avg)	`query_graph` with `path(from, to)`	Single call, ~$0.01, deterministic chain

8. Methodology

8.1 Experiment Infrastructure

All experiments share a common infrastructure:

- **Dummy Terraform project:** 75 `null_resource` instances across 6 modules (networking, security, compute, database, loadbalancer, monitoring) simulating a 3-tier AWS deployment. Produces 4,041 lines of state JSON (~33K tokens). Zero cloud charges.
- **10 task prompts** spanning 3 difficulty levels and 9 categories, with ground truth answers and 4 scoring types
- **Automated pipeline:** `runner.ts` -> `scorer.ts` -> `visualize.ts`

8.2 Experiment 1: Tool-Augmented vs. Raw State

Design: Same 10 prompts, same infrastructure. Two arms:

- **Raw arm (complete):** Agent has only Bash tool, runs `terraform state list`, `terraform show -json`, etc.
- **MCP arm (pending v1):** Agent has 12 MCP tools: `list_resources`, `get_dependencies`, `query_graph`, etc.

Isolation: Each trial runs in a temporary directory with only `.tf` files copied in (no README, no prompts.json). The `.claude/` directory is deleted between trials to prevent cross-trial context leakage.

8.3 Experiment 2: Abstraction Granularity

Design: Same prompts, three tool granularity levels:

- **Coarse:** Tools return full resource configurations (all attributes)
- **Medium:** Tools return summaries + key attributes only
- **Fine:** Tools return minimal fields (ID, type, summary string)

Tradeoff evaluated: Fewer tokens vs. risk of insufficient context for complex queries.

8.4 Experiment 3: Cross-Model Portability

Design: Same 10 prompts through the MCP server, connected to 15 models across proprietary and open-source providers. All selected models have native tool-calling / function-calling support.

Proprietary Models (9):

Model	Provider	Why Included
Claude Sonnet 4.6	Anthropic	Baseline model, native MCP support (Feb 2026)
Claude Haiku 4.5	Anthropic	Budget tier, tests if MCP tools compensate for smaller model
GPT-5.4	OpenAI	Latest OpenAI flagship, complex reasoning + coding
GPT-5.4 mini	OpenAI	Cost-optimized, lower latency variant
o3	OpenAI	Reasoning model, math/science/coding focus
Gemini 2.5 Pro	Google	Latest Gemini, extended context + tool use
Gemini 2.5 Flash	Google	Fast/cheap workhorse, advanced reasoning
Grok 4.1 Fast	xAI	Best agentic tool calling, 2M context window
MiniMax-M2.7	MiniMax	1M context, agentic productivity and multi-step planning

Open-Source Models (6):

Model	Provider	Why Included
Qwen3.5-397B-A17B	Alibaba	Latest flagship MoE, multimodal reasoning + tool calling
DeepSeek-V3.2	DeepSeek	Strong reasoning, IMO/IOI gold medal results, agentic tool-use pipeline
Llama 4 Maverick (17B active)	Meta	MoE architecture, native tool calling
GLM-4.5-Air	THUDM/Zhipu	Optimized for tool use and web browsing
MiMo-V2-Flash	Xiaomi	Trained explicitly for agentic and tool-calling workflows
Mistral Medium 3	Mistral AI	Native function calling without special prompting

Tradeoff evaluated: Whether the MCP abstraction is model-agnostic or biased toward specific providers. If smaller/cheaper models (Haiku, GPT-5.4 mini, Gemini Flash, MiMo-V2-Flash) achieve comparable accuracy to flagship models when given structured MCP tools, it validates that **tool design matters more than model size** for infrastructure tasks.

8.5 Scoring Framework

Type	Used For	Scoring
`substring-match`	Attribute lookups (exact values)	>=100% found -> 1.0, >=50% -> 0.5, else 0.0
`set-overlap`	Enumeration, dependency lists	>=80% matched -> 1.0, >=50% -> 0.5, else 0.0
`topological-validation`	Deployment order, dependency chains	Nodes found (50%) + precedence rules (50%), threshold 0.8
`checklist`	Planning, inventory prompts	>=min_for_full -> 1.0, >=min_for_partial -> 0.5

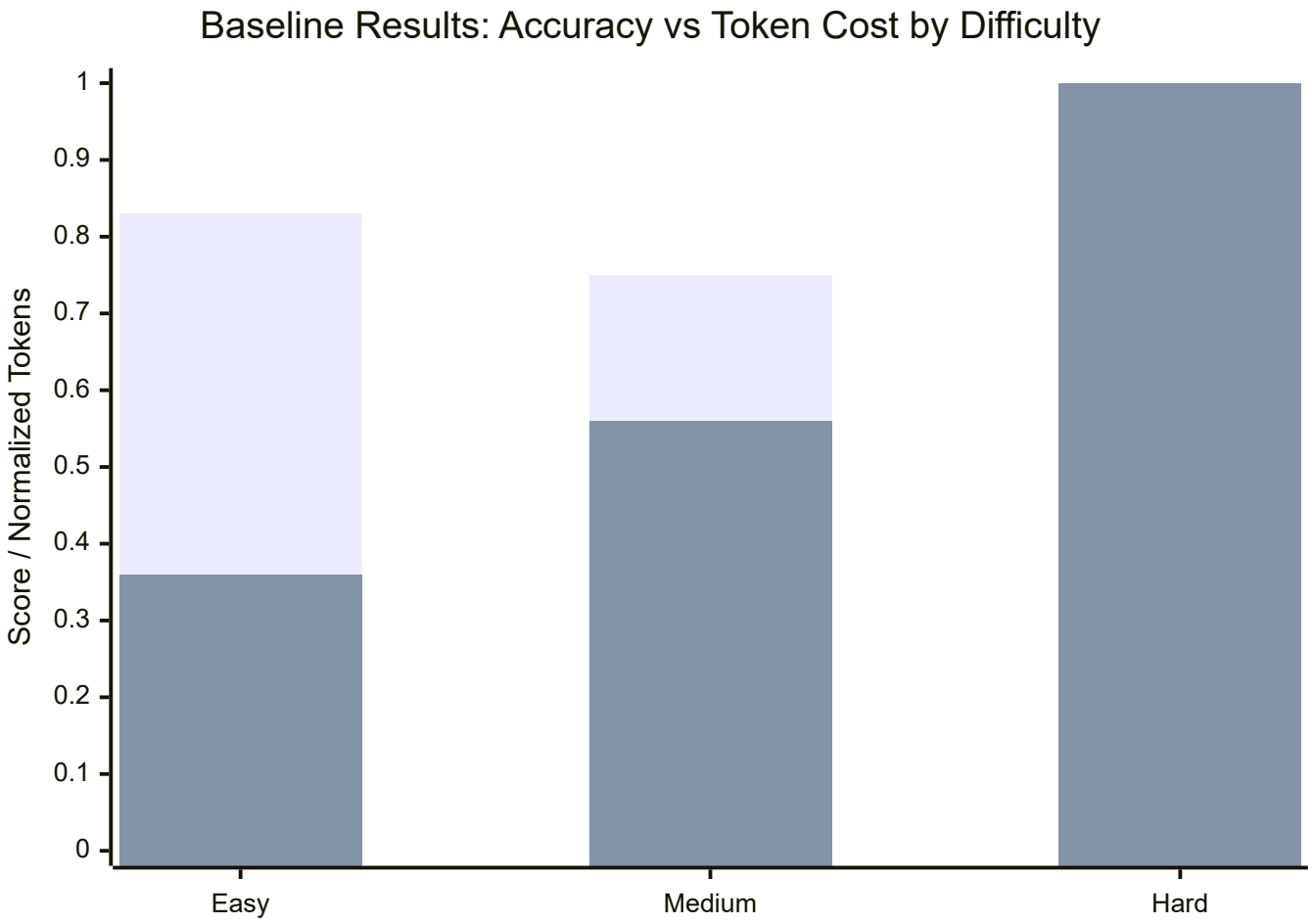
Fuzzy matching includes AZ short-code mapping (`us-east-1a` to `use1-az1`), separator-agnostic number matching, and compact form normalization.

9. Preliminary Results

9.1 Baseline Experiment (Experiment 1, Raw Arm)

Metadata: 10 prompts x 3 trials = 30 API calls. Claude Sonnet via CLI. Total: 1,491,955 tokens, $1.73, 696.7 seconds.

Difficulty	Accuracy	Avg Tokens	Avg Cost	Avg Tool Calls	Tokens/Correct Answer
Easy	0.83	28,219	$0.028	2	28,157
Medium	0.75	44,228	$0.047	3	43,094
Hard	**0.94**	78,584	$0.102	6	83,164
Overall	**0.83**	**49,732**	**$0.058**	**3.5**	--



9.2 Key Findings

Finding 1: Hard prompts maintain high accuracy but at 2.8x token cost.

	Easy	Hard	Ratio
Accuracy	0.83	0.94	1.1x
Tokens	28K	79K	**2.8x**
Cost	$0.028	$0.102	**3.6x**
Tool calls	2	6	**3x**
Tool output	5K chars	34K chars	**6.9x**

The initial hypothesis predicted hard accuracy at 10-30%. Actual: **94%**. The problem at 75-resource scale is not accuracy degradation, it is **cost and latency explosion**.

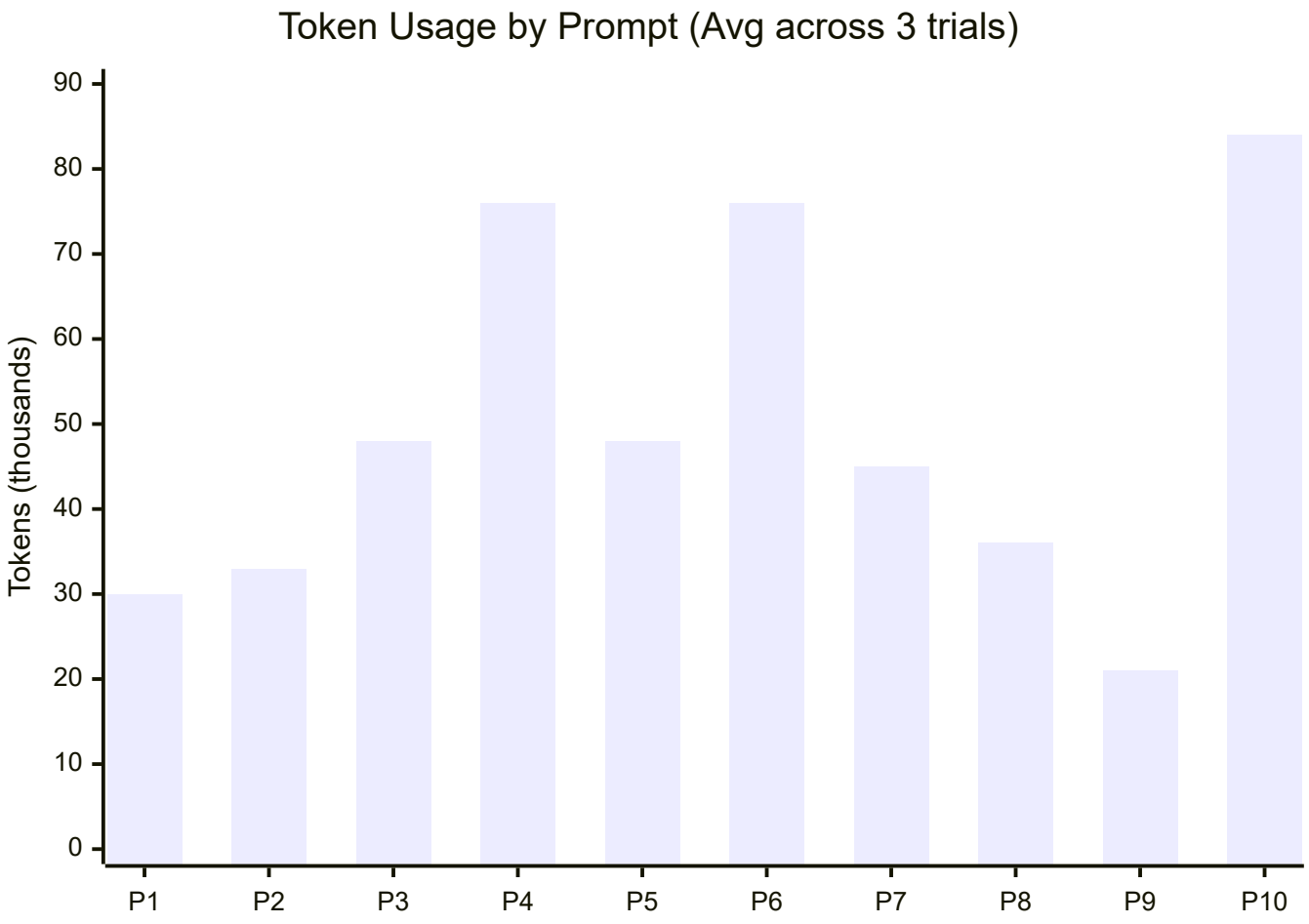
Finding 2: 97.9% of all tokens are input (context accumulation).

> Input tokens / Total tokens = 1,460,877 / 1,491,955 = **97.9%**

As Claude makes more tool calls, the accumulated conversation history (with all prior tool outputs) is re-sent with each API call. This is the core inefficiency MCP tools address: by returning pre-filtered, smaller responses, each turn accumulates less context.

Finding 3: Token variance is high across trials for the same prompt.

Prompt 4 (impact analysis) scored 1.0 on all 3 trials, but token usage ranged from 51K to 99K, a **1.9x variance** for identical results. This non-deterministic cost is problematic for production use and is eliminated by deterministic MCP tool responses.



9.3 Projected MCP Improvement

Metric	Raw CLI (actual)	MCP (projected)	Reduction
Avg tokens/prompt	49,732	~8,000	**~6x fewer**
Avg tool calls	3.5	1.2	**~3x fewer**
Avg cost/prompt	$0.058	~$0.012	**~5x cheaper**
Token variance	High (2-3x)	Low (deterministic)	Stable

9.4 What Remains

Item	Depends On	Timeline
Experiment 1, MCP arm	v1 implementation (Phases 1-7)	Apr 1-7
Experiment 2, Granularity	v1 + tool response variants	Apr 7-9 (if time permits)
Experiment 3, Cross-model	v1 + GPT/Gemini API integration	Future work
Scorer refinements	Prompt 3 failure analysis	Apr 1-2

Worst-case workload: If v1 implementation takes longer than expected, Experiments 2 and 3 will be descoped. Experiment 1 alone (raw vs MCP comparison) validates the core thesis. Experiments 2 and 3 become concrete future work.

Base-case workload: v1 completes by Apr 5, Experiment 1 MCP arm runs Apr 6-7, Experiment 2 runs Apr 8-9, report finalized Apr 10.

10. Impact

10.1 Who Cares

- **DevOps teams:** At 100 agent queries/day, MCP tools save ~$1,679/year per project on token costs alone. At enterprise scale (10+ projects), savings multiply.
- **Platform engineers:** Deterministic, filtered tool responses mean agents produce reliable infrastructure changes, not hallucinated configurations.
- **Terraform community:** An open-source MCP server that any Terraform user can install and connect from Claude Desktop, Claude Code, Cursor, or any MCP-compliant client.
- **LLM tool developers:** Our experiment data provides concrete evidence for how purpose-built tools vs. raw CLI access affects token cost, accuracy, and latency, applicable beyond Terraform.

10.2 Classmate Usability

Anyone with **Terraform CLI + Node.js** can install and test the MCP server:

```
git clone https://github.com/KalharPandya/terraform-smart-context-mcp-42
cd terraform-smart-context-mcp-42 && npm install
cd experiments/baseline/dummy-infra && terraform init && terraform apply -auto-approve
```

The dummy infrastructure uses only `null_resource`, **zero cloud charges**, instant apply. Connect the MCP server to Claude Desktop or Cursor and query the 75-resource infrastructure through purpose-built tools.

10.3 Broader Significance

This is one of the first MCP servers that provides structured access to **live infrastructure state**, not just documentation. HashiCorp's MCP server answers "what does this provider support?" Ours answers "what is actually deployed?" If the experiment results confirm the projected 6x token reduction, the approach generalizes to any domain where LLMs interact with large structured state (databases, Kubernetes clusters, cloud consoles).